

CORRIGENDUM NO. 6

EXTRA-LOW VOLTAGE SYSTEM

APPENDIX N SMART TECHNOLOGIES BUILT IN IBMS/ BMS

The Nominated Sub-Contractor shall provide all the necessary hardware, software and field devices as complete solutions to achieve the required functions.

<u>S/N</u>	<u>Smart Technologies</u>	<u>Functions</u>	<u>Base/Optional</u>	<u>Description</u>
<u>1</u>	<u>Smart chiller plant optimization</u>	<u>Optimize chiller plant equipment (chillers, pumps & cooling towers) operation points to maximize chiller plant efficiency</u>	<u>Base</u>	• The BMS/ iBMS shall optimize the operation of chillers, pumps, and cooling towers to achieve maximum efficiency while maintaining desired environmental conditions. • The system shall integrate with all chiller plant equipment and utilize real-time data analytics to adjust operational setpoints dynamically. • The BMS/ iBMS shall support open communication protocols such as BACnet, Modbus, and LonWorks to ensure seamless integration with third-party equipment. • The system shall provide automated control, monitoring, reporting, and alarms for all chiller plant components. • The BMS/ iBMS shall continuously monitor chiller parameters, including Chilled water supply and return temperature; Chilled water flow rate; Compressor loading and energy consumption (kW/Ton); Condenser water supply and return temperature; and Chiller runtime and cycling frequency. • The system shall implement adaptive load management to: Optimize chiller sequencing to match cooling demand; Distribute loads evenly across multiple chillers to maximize efficiency; and prevent excessive cycling and unnecessary startup of additional chillers. • Implement chilled water setpoint optimization using outdoor temperature and cooling load demand. • Support predictive maintenance alerts based on chiller efficiency trends and operational anomalies. • The system shall provide real-time monitoring and control of primary and secondary chilled water pumps, including differential pressure across chilled water loop; pump speed and power consumption; and flow rate and temperature differential. • Variable Frequency Drives (VFDs) shall be controlled dynamically to: Maintain the optimal differential pressure setpoint; Reduce pump energy consumption while ensuring adequate cooling supply; and minimize unnecessary pump runtime and staging. • Implement an automated pump staging sequence to ensure efficient operation based on system load. • The system shall monitor and control cooling tower parameters, including Condenser water supply and return temperature; Fan speed and energy consumption; Wet bulb temperature and approach temperature; and water flow rate and conductivity for water treatment management. • Optimize cooling tower fan speed using VFDs to: Maintain optimal approach temperature for efficient heat rejection; and reduce excessive energy consumption by adjusting fan speeds dynamically. • Implement condenser water temperature setpoint optimization based on ambient wet bulb temperature and chiller performance characteristics. • The BMS/ iBMS shall integrate with energy meters, temperature sensors, and flow meters for real-time performance analysis. • AI-based analytics shall be incorporated for predictive optimization and fault detection. • Support web-based remote monitoring and control capabilities.

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2	<u>Predictive maintenance for M&E major equipment serving critical hospital services</u>	<u>Early detection of potential equipment failure and schedule the critical and normal maintenance in time</u>	<u>Base</u>	• This function is designed to ensure early detection of potential equipment failures and schedule timely maintenance for major mechanical and electrical (M&E) assets that support critical hospital services. It employs advanced sensor technologies, machine learning analytics, and integration with maintenance management systems to proactively address issues. • Sensor Integration ▪ Types: IoT sensors (vibration, temperature, pressure, flow, acoustic) installed on critical assets such as HVAC systems (including OTs and isolation wards), generators, UPS systems, and life-support equipment. ▪ Data Granularity: Continuous real-time monitoring with data capture intervals as low as 1 minute for high-priority assets, aggregating to 15-minute averages for trend analysis. • Data Acquisition & Communication ▪ Protocols: Utilizes industry-standard protocols (BACnet, Modbus, MQTT) for seamless data collection from the installed sensors. ▪ Connectivity: Secure wired/wireless network connections with redundant pathways to guarantee data integrity in the hospital's critical zones. • Analytics & Machine Learning ▪ Predictive Algorithms: Deploy machine learning models (e.g., regression analysis, anomaly detection, time-series forecasting) trained on historical performance and failure data to predict potential malfunctions. ▪ Threshold Analysis: Defines operational thresholds based on manufacturer commissioning data and historical performance benchmarks; deviations trigger early alerts. • Integration with Maintenance Systems ▪ CMMS/FMS Integration: Automated creation of maintenance work orders through integration with the hospital's Computerized Maintenance Management System (CMMS) when alerts indicate potential failures. ▪ Priority Scheduling: Differentiates between critical and normal maintenance tasks to prioritize interventions based on risk assessments and patient safety impact. • Visualization & Dashboarding ▪ Real-Time Dashboards: Centralized dashboards display asset health indicators, predictive maintenance forecasts, alert statuses, and historical trends. ▪ Alarm & Notification System: Configurable alerts and SMS/email notifications directed to technical teams upon detection of abnormal sensor readings or predictive model warnings. • Data Storage & Historical Analysis ▪ Data Lake: Storage of both real-time and historical sensor data to support continuous model re-training, trend analysis, and compliance reporting. ▪ Reporting: Generate periodic predictive maintenance reports that compare actual asset performance against predicted trends to validate model accuracy and equipment reliability.
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133	<u>Smart UPS monitoring system for both centralized UPS system and localized UPS system in LAN rooms</u>	- <u>Realtime monitoring of UPS performance and battery health;</u> - <u>Early detection and alarm when there is abnormal battery temperature rise;</u>	<u>Base</u>	• This function provides centralized and localized monitoring of Uninterruptible Power Supply (UPS) systems across the building, particularly for critical areas such as LAN rooms. It ensures reliable UPS performance by continuously monitoring key parameters — including battery health, load, and temperature — and triggering early alerts when abnormal conditions, such as excessive battery temperature rise, are detected. • Real-Time Performance Monitoring ▪ Data Acquisition: Continuous monitoring of UPS metrics (battery voltage, current, capacity, load percentage, temperature, and runtime). ▪ Sampling Frequency: Typically, data captured every 1-5 minutes for critical parameters to ensure early anomaly detection. • Battery Health Monitoring ▪ Health Metrics: Tracks battery state-of-charge (SOC), depth of discharge, cycle count, and overall battery condition. ▪ Historical Trending: Aggregates and compares current battery performance with historical baseline data to assess degradation over time. • Abnormal Temperature Detection ▪ Temperature Sensors: Integrated sensors measure battery temperature in real time. ▪ Threshold Settings: Configurable thresholds for allowable temperature rise based on manufacturer specifications and environmental conditions; triggers alarms when thresholds are exceeded. • Alarm & Notification System ▪ Automated Alerts: Immediate alarms via the BMS dashboard, SMS, email, or local audible alarms when abnormal battery temperature or performance issues are detected. ▪ Escalation Protocols: Integration with CMMS for automatic work order generation and timely technician notification for remediation. • Integration & Data Communication ▪ Protocol Compatibility: Uses standard communication protocols (BACnet, Modbus, MQTT, RESTful APIs) for seamless integration with existing BMS/EMS. ▪ Centralized Data Aggregation: Supports both centralized monitoring (for the overall UPS system) and localized monitoring (within LAN rooms) in a single unified dashboard. • User Interface & Dashboard ▪ Visualisation: Real-time dashboards displaying key UPS performance indicators, battery health status, and temperature trends. ▪ Customizable Views: Role-based display options for facility managers, IT administrators, and operations teams.
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4	<u>Energy optimization</u>	<u>Automated Energy and Water Management.</u> <u>Monitor building performance and ensure occupant comfort. For example:</u> - IoT sensors to control aircon airside system. <u>Uses occupancy sensors, climate data, and advanced algorithms to adjust heating, cooling, and ventilation;</u> - Remote Operation to allow both local and remote control of lighting systems, enhancing accessibility and usability; - Sensor based operations that supports sensor based operations for automated responses to occupancy and ambient light levels, optimizing energy efficiency.	<u>Optional</u>	• This function focuses on automated energy and water management to improve building performance and ensure occupant comfort while optimizing energy efficiency. It leverages IoT sensors, remote control systems, and sensor-driven operations to automatically adjust HVAC, lighting, and related systems based on real-time occupancy and ambient conditions. • IoT Sensor Network ▪ Deploy sensors to capture data for: - o Occupancy: Infrared, thermal, or camera-based sensors to detect presence and density of occupants. - o Climate Data: Temperature, humidity, and CO₂ sensors for air quality monitoring. - o Ambient Light: Lux sensors for lighting control. • Airside Control Integration ▪ Connect with the air conditioning (AC) system's airside controls to: - o Adjust airflow and dampers dynamically in response to occupancy changes. - o Regulate air temperatures and humidity based on sensor feedback and external climate data. - o Optimize ventilation schedules to preserve energy and maintain comfort. • Advanced Algorithmic Control ▪ Use machine learning and rule-based algorithms to predict building usage patterns and adjust: - o HVAC setpoints based on historical and real-time data. - o Lighting levels by integrating occupancy and ambient light sensor data. - o Adaptive scheduling modes for peak and off-peak periods. • Remote Operation Capability ▪ Enable both local and remote control of: - o Lighting systems through secure web interfaces. - o HVAC and other building automation systems via integrated platforms. - o User dashboards that display energy and water usage, occupancy levels, and system performance. • Automated Response & Control ▪ Implement sensor-based operations that support: - o Immediate response to occupancy changes (e.g., activating or dimming lights, adjusting HVAC settings). - o Fault detection and automated resets based on thresholds. - o Integration with predictive analytics to continuously fine-tune energy and water usage. • Energy and Water Data Aggregation ▪ Continuous collection and aggregation of data from energy meters, water meters, and sensor arrays with: - o 15-minute to hourly data granularity for trend analysis. - o Baseline and post-intervention comparisons to support energy-saving initiatives. - o Data export capabilities for ESG reporting.
5	<u>Performance qualification</u>	<u>Monitor the asset performance to verify that operations are operating efficiently.</u> - Continually monitor that assets are operated as intended; - Compare data with original commissioning figures to confirm if equipment is operating as per design intent; - Benchmarking & ACMV Optimization.	<u>Optional</u>	• The Performance Qualification function continuously monitors asset operation to confirm that equipment is running as designed. It compares live operational data with original commissioning parameters and benchmarks asset performance to identify opportunities for optimization, particularly within the ACMV systems. • Continuous Operational Monitoring ▪ Data Acquisition: Collects real-time operational parameters (e.g., supply/return air temperatures, airflow rates, fan speeds, compressor cycles, power consumption) from BMS sensors and IoT devices at intervals ranging from 15 minutes to hourly. ▪ Real-Time Verification: Monitors key variables to ensure assets (chillers, AHUs, pumps, etc.) are operating within predefined design thresholds. • Commissioning Data Comparison

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				▪ Baseline Data Repository: Stores original commissioning figures, including design setpoints, airflow capacities, thermal performance data, and energy ratings for each asset. ▪ Automated Analysis: Employs algorithms to compare current performance metrics with baseline commissioning data, highlighting deviations through percentage differences and trend deviations. • Benchmarking & ACMV Optimization ▪ Key Performance Indicators (KPIs): Defines and tracks specific KPIs, such as: - o Chiller efficiency (kW/RT) - o Energy Utilisation Index (EUI) for ACMV systems (kWh/m²/year) - o Equipment runtime and cycling frequency ▪ Trend Analysis: Utilizes historical data and regression techniques to benchmark current performance against historical performance averages and best-practice standards. ▪ Optimization Alerts: Triggers notifications when asset performance deviates significantly from the commissioned performance, suggesting the need for optimization or maintenance. • Visualization & Reporting ▪ Dashboard Integration: Provides interactive dashboards that display real-time asset performance, historical trends, and discrepancies between actual and commissioned performance metrics. ▪ Automated Reporting: Generates regular reports (monthly/quarterly) that summarize performance verification, highlight performance gaps, and provide actionable insights for ACMV optimization. • Predictive Analytics & Anomaly Detection ▪ Machine Learning Models: Deploys predictive models to detect early signs of performance degradation or operational anomalies. ▪ Maintenance Integration: Interfaces with CMMS/FMS systems to automatically generate maintenance tickets when deviations exceed acceptable thresholds, thereby facilitating proactive interventions. • User Alerts & Notifications ▪ Threshold-based Alerts: Configurable alerts for operations teams when real-time data indicates that asset performance falls outside expected parameters. ▪ Audit Trail: Maintains an audit log of comparisons, alerts, and corrective actions for accountability and continuous improvement.
6	<u>Bench marking</u>	- <u>Breakdown energy loads into various sub-categories:</u> - <u>Assess what buildings may have excessive loads (i.e. – lighting, lifts, ACMV).</u>	<u>Base</u>	• This function analyzes and benchmarks energy consumption data by breaking down overall usage into various sub-categories such as lighting, lifts, and ACMV systems. It provides insights into specific building subsystems to identify inefficiencies and excessive loads. The goal is to inform stakeholders and facility managers about energy performance relative to predetermined baselines and industry standards. • Sub-metering and Data Collection ▪ Data Sources: Integration with sub-meters installed on lighting, lifts, ACMV systems, and other relevant energy-consuming systems. ▪ Data Frequency: Collection at intervals (e.g., 15 minutes or hourly) to capture detailed energy profiles. ▪ Communication Protocols: Supports BACnet, Modbus, MQTT, and RESTful APIs. • Energy Load Breakdown ▪ Categorization: Automatically classify total building energy consumption into distinct sub-categories: - o Lighting - o Lifts - o ACMV (Air Conditioning & Mechanical Ventilation) - o Other critical systems.

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				• Benchmarking & Comparative Analysis ▪ Baseline Setting: Utilize historical data and commissioning figures to establish baseline energy usage for each category. ▪ Benchmark Metrics: Apply KPIs such as energy intensity (kWh/m²), system-specific load factors, and percentage shares of total load. ▪ Comparative Analysis: Compare current usage against established benchmarks and peer data. • Excessive Load Detection ▪ Threshold Evaluation: Define thresholds for acceptable energy consumption based on design intent and industry best practices. ▪ Anomaly Detection: Use statistical or machine learning models to detect abnormal energy consumption patterns (e.g., excessive lighting or ACMV loads). • Visualization & Dashboarding ▪ Interactive Dashboards: Present detailed graphical views that illustrate energy consumption breakdown, temporal trends, and load comparisons across different building areas or systems. ▪ Custom Reports: Generate automated reports to highlight excessive loads and performance deviations, available in PDF, Excel, or online dashboards. • Alerting & Reporting ▪ Automated Alerts: Configure notifications for facility managers when energy loads exceed predefined thresholds for any sub-category. ▪ Periodic Reporting: Produce scheduled reports (daily, weekly, monthly) summarizing energy breakdown, benchmarking outcomes, and recommendations for load optimization.
7a	One stop shop software platform	<u>Collect and visualize facility management data.</u> <u>Data-driven asset maintenance involves IoT-enabled sensor technology that connects critical assets across essential infrastructure, generating insights into operating conditions such as humidity, vibration, and temperature.</u>	Optional	• This platform serves as a centralized hub for collecting, aggregating, and visualizing facility management data from a variety of IoT-enabled sensors. It connects with critical assets across essential infrastructure to monitor operating conditions — such as humidity, vibration, and temperature — and supports proactive, data-driven asset maintenance. The system enhances situational awareness and enables efficient decision-making, aligning with Singapore's Green Mark and Smart Nation initiatives. • Data Ingestion & Aggregation ▪ Supports multiple data sources including IoT sensors, BMS, and legacy systems. ▪ Utilizes standard protocols (BACnet, Modbus, MQTT, RESTful APIs) for connectivity. ▪ Aggregates real-time and historical data into a centralized data lake with data normalization and quality checks. • Sensor Technologies ▪ IoT sensors for environmental and operational parameters: humidity, vibration, temperature, and others. ▪ Configurable sensor calibration and fault detection to maintain data accuracy. ▪ Wireless and wired sensor integration for flexible deployment. • Real-Time Visualization ▪ Interactive dashboards with customizable widgets and charts (line graphs, heatmaps, gauges, etc.). ▪ Visual summary of critical asset performance, trends, and anomalies. ▪ Geospatial mapping to visualize asset locations across facilities. • Data-Driven Asset Maintenance ▪ Predictive maintenance alerts generated by integrating sensor data with historical trends. ▪ Automated rule-based notifications when predefined thresholds (e.g., excessive vibration or abnormal temperature fluctuations) are breached. ▪ Integration with CMMS for automated work order generation and scheduling. • Analytics & Insights ▪ Advanced analytics engine utilizing machine learning or statistical models for fault

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				- prediction and performance optimization. - Customizable reports to track asset health, system performance, and energy efficiency. - Support for digital twin models to simulate and optimize facility operations. - User Interface & Accessibility - Web-based dashboard accessible via modern browsers and mobile applications. - Role-based access control (RBAC) to ensure data security and compliance with PDPA. - Export capabilities in PDF, Excel, and other common formats for reporting and audit trails.
	<u>Data-Driven maintenance management system</u>	<u>Automate workflow processes across systems. Combined with control software, advanced analytics to inform and enable predictive and condition-based maintenance programs.</u>	<u>Optional</u>	- This function automates maintenance workflow processes by integrating data from various building systems, sensors, and control platforms. Through advanced analytics and predictive algorithms, it transforms raw operational data into actionable insights that drive condition-based maintenance and predictive maintenance programs. - Automated Workflow Processes - Workflow Orchestration: Automates maintenance processes by routing data-driven alerts to relevant personnel. - Event-Driven Triggers: Initiates maintenance tasks automatically based on predefined conditions (e.g., sensor anomalies, performance deviations). - Integration with CMMS/ FMS: Seamlessly pushes work orders and maintenance schedules into existing systems. - Data Integration & Aggregation - Multi-Source Data Collection: Ingests data from BMS, IoT sensors, control software, and legacy systems using standard protocols (BACnet, Modbus, MQTT, RESTful APIs). - Data Normalization: Aggregates real-time and historical data into a centralized data lake for analysis. - Advanced Analytics & Predictive Algorithms - Predictive Maintenance Models: Utilizes machine learning and statistical techniques to forecast asset failures and recommend optimal maintenance timing. - Condition-Based Monitoring: Continuously monitors asset parameters (e.g., vibration, temperature, power consumption) to assess equipment health. - Anomaly Detection: Detects irregularities and deviations from normal performance to trigger alerts. - Control Software Integration - Automated Controls: Integrates with control systems to execute predefined corrective actions remotely (e.g., system resets, load shedding). - Feedback Loop: Uses closed-loop controls to validate that maintenance actions have resolved the detected issue. - Reporting & Dashboarding - Interactive Dashboards: Visualize maintenance KPIs, asset health trends, work order statuses, and predictive analytics outcomes. - Automated Reports: Generates periodic performance, maintenance, and compliance reports tailored for stakeholders. - Notification & Alert System - Configurable Alerts: Triggers notifications via SMS, email, or in-dashboard alerts when anomalies are detected, or thresholds are exceeded. - Escalation Procedures: Implements tiered alert systems and escalation paths for unresolved issues. - Security & Audit Logging - Audit Trail: Maintains comprehensive logs of all maintenance activities, alerts, and automated actions for review and compliance.
	<u>Data ingestion</u>	<u>- Consolidate asset management strategies via integration with Computerized Maintenance Management Systems (CMMS).</u>	<u>Optional</u>	The Data Ingestion function gathers and consolidates data from various sources related to asset management, energy usage, and predictive maintenance. This includes seamless integration with Computerized Maintenance Management Systems (CMMS) and the

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		- <u>Integrate with iHAM.</u> - <u>Predictive maintenance: AI algorithms to analyze data from equipment to predict failures and schedule maintenance proactively.</u> - <u>Energy optimization: Machine learning models can analyze historical energy usage data and weather forecasts to suggest ways to reduce energy consumption.</u>		Integrated Health Asset Management (iHAM) platform, enabling comprehensive asset oversight. Advanced AI and machine learning algorithms are applied to the ingested data to predict maintenance needs and optimize energy consumption. • CMMS Integration ▪ Data Source: Connects with established Computerized Maintenance Management Systems (CMMS) to import work orders, asset histories, and maintenance schedules. ▪ Protocols: Supports RESTful APIs, SOAP, or custom data feeds provided by the CMMS vendor. • iHAM Integration ▪ Data Convergence: Facilitates integration with the iHAM (Integrated Health Asset Management) platform to consolidate healthcare-related asset data, maintenance logs, and compliance records. ▪ Standard Interfaces: Utilizes standard data exchange formats (e.g., XML, JSON). • Predictive Maintenance Analytics ▪ AI Algorithms: Deploys machine learning models (e.g., time-series forecasting, anomaly detection) trained on historical equipment performance and failure data to predict potential issues and proactively schedule maintenance. ▪ Data Inputs: Sensor data from equipment, maintenance logs, and environmental conditions. • Energy Optimization Analytics ▪ Machine Learning Models: Uses algorithms to analyze historical energy usage patterns and correlate them with weather forecast data. ▪ Output: Generates actionable insights and recommendations for reducing energy consumption and cost, optimizing system setpoints and operating schedules. • Data Normalization & Storage ▪ Centralized Data Lake: Aggregates data from CMMS, iHAM, sensor networks, and external sources into a unified repository with standard data formats for ease of analysis. ▪ Data Normalization: Applies ETL (Extract, Transform, Load) processes to ensure data consistency and quality. • Data Security & Compliance ▪ Access Controls: Enforces role-based access (RBAC) and audit logging across the data ingestion pipeline. • Real-Time & Batch Data Processing ▪ Operational Modes: Supports both real-time data streaming (using MQTT, WebSocket protocols) and scheduled batch updates to accommodate various operational requirements and data volumes. • Visualization & Reporting ▪ Dashboard Integration: Feeds aggregated and processed data into dashboards that display key metrics for asset health, maintenance trends, and energy usage analytics. ▪ Report Generation: Automates the production of periodic reports to support decision-making and regulatory compliance.
	<u>Review & Track</u>	<u>Set and track operational KPI goals. Monitor asset health, pre-empt and predict asset failure, including dynamic end-of-life estimation.</u> - <u>Defect closeout rate</u> - <u>No. of issues identified per equipment</u> - <u>Identified savings</u> - <u>Energy Benchmarking & Carbon Reporting</u>	<u>Optional</u>	• The Review & Track function is aimed at providing a comprehensive framework to monitor operational KPIs, assess asset performance, and predict potential failures through data analytics. The function includes dynamic estimation of asset end-of-life (EOL) and continuous tracking of key performance indicators such as the defect closeout rate, issues per equipment, identified savings, as well as energy benchmarking and carbon reporting. • KPI Goal Setting & Tracking ▪ Definable KPIs: Facility managers can define KPIs such as defect closeout rate, number of issues per equipment, identified cost/energy savings, and carbon intensity metrics (kgCO₂e/m²/year). ▪ Dashboard Monitoring: Real-time dashboards display KPI trends, targets vs. actuals, and dynamic performance indicators.

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				▪ Custom Alerts: Configure alert thresholds when KPIs deviate from set goals. • Asset Health Monitoring ▪ Continuous Data Acquisition: Ingest real-time data from asset sensors (vibration, temperature, pressure, power consumption) to evaluate equipment condition. ▪ Health Index Calculation: Derive an asset health index using machine learning algorithms and rule-based models calibrated against commissioning data and historical performance. • Predictive Maintenance & Dynamic EOL Estimation ▪ Predictive Analytics: Apply AI-driven models (e.g., time-series analysis, regression, anomaly detection) to forecast equipment failure likelihood. ▪ Dynamic EOL Calculation: Continuously update the expected end-of-life of critical assets based on operational stress, maintenance history, and degradation patterns. ▪ Automated Workflows: Integrate with CMMS/FMS for scheduling preventive maintenance and repair actions as predicted by analytics. • Defect Closeout & Issue Identification ▪ Defect Closeout Rate: Track and calculate the ratio of resolved defects versus reported issues over a defined period. ▪ Issues Per Equipment: Log and analyze the frequency and type of issues per asset, facilitating root-cause analysis. ▪ Reporting Tools: Generate periodic reports summarizing defect resolution rates and outstanding issues. • Identified Savings & Benchmarking ▪ Energy & Cost Savings: Quantify savings derived from operational improvements and maintenance optimizations, displaying both monetary and energy consumption impacts. ▪ Benchmarking: Use sub-metered and historical data to benchmark each building's performance against established baselines and peer data. • Energy Benchmarking & Carbon Reporting ▪ Energy KPI Dashboards: Visualize energy utilisation indices (EUI) and correlate with operational improvements and savings. ▪ Carbon Reporting: Automatically calculate carbon emissions using local emission factors (e.g., NEA grid factors) and report on performance relative to targets. ▪ Compliance Reporting: Support compliance with BCA Green Mark and ESG reporting requirements by generating exportable detailed reports.
	Automated fault detection	<u>Utilize Switch analytics tools to automatically detect system inefficiencies</u> - Identify system faults - Triage maintenance through identification of highest priority faults	Optional	• The Automated Fault Detection function leverages fault detection and diagnostics (FDD) analytics — often referred to as Switch analytics tools or similar solutions—to automatically detect inefficiencies and faults in building systems. These tools are designed to integrate with Building Management Systems (BMS), prioritize faults based on impact, and triage issues to streamline maintenance workflows. • Real-Time Fault Detection ▪ Continuous monitoring of system data streams (e.g., temperature, humidity, valve position, setpoints, flow rates, energy consumption) from BMS. ▪ Detect anomalies using predefined rules (ASHRAE-based FDD libraries) and AI-driven pattern recognition models. • Fault Classification ▪ Categorizes faults by system (e.g., AHU, chiller, VAV box, pump) and fault type (e.g., sensor drift, valve stuck open, simultaneous heating/cooling). • Priority-Based Triage ▪ Assigns fault severity scores based on operational impact, energy loss potential, or risk to occupant comfort and safety. ▪ Enables prioritization of urgent maintenance activities. • Switch or FDD Analytics Engine ▪ Integrates with third-party FDD platforms or uses native BMS analytical tools.

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				▪ Rule sets configurable to local equipment and building typologies. • Root Cause Identification ▪ Uses rule-based and data-driven diagnostics to trace symptoms back to root causes (e.g., sensor calibration errors, equipment mismatch, poor control logic). • Maintenance Recommendations ▪ Automatically generates actionable insights and recommended corrective actions. ▪ Supports integration with CMMS/FMS platforms for work order creation. • Visualization & Reporting ▪ Dashboards display current faults, severity levels, trends over time, and maintenance outcomes. ▪ Weekly/monthly fault summary reports for management review. • Performance Tracking ▪ Monitors repeat fault patterns and tracks fault resolution time and defect closeout rates. ▪ Benchmarks building/system fault rates against industry norms. • Integration Points ▪ BMS Systems: Real-time data feeds and control system context (BACnet/IP, Modbus). ▪ CMMS/iBMS Platforms: Push fault logs and recommendations into maintenance workflows. • Alerting & Notification ▪ Configurable alert settings via email, SMS, or dashboard pop-ups when critical faults are detected. ▪ Escalation logic for unresolved or persistent issues.
7b	<u>Data visualization</u>			
	<u>Asset management</u>	<u>Access high-level data such as EUI, or Carbon Intensity</u>	<u>Optional</u>	• The Data Visualization – Asset Management function enables facilities and sustainability managers to access high-level performance indicators across assets in real-time. Key metrics such as Energy Utilisation Index (EUI) and Carbon Intensity are displayed in an intuitive visual interface. • KPI Dashboard Interface ▪ Real-time visual dashboard displaying key metrics including: - o EUI (kWh/m²/year) - o Carbon Intensity (kgCO₂e/m²/year) - o Water Use Intensity (L/m²/year) - o Waste Diversion Rate (%) - o Chiller Plant Efficiency (kW/RT) • Hierarchical Asset View ▪ Organize and display assets by type (e.g., chillers, AHUs, lighting systems), zone, floor, or building cluster. ▪ Enables multi-building or portfolio-level overview. • Interactive Charts & Heatmaps ▪ Time-series visualizations for each KPI by hour, day, month, and year. ▪ Custom heatmaps by zone to highlight underperforming areas or assets. • Drill-Down Capability ▪ Click-through interface allows users to view underlying system or equipment performance behind high-level KPIs. ▪ Includes trend comparisons with baselines and targets. • Carbon Intensity Calculations ▪ Computed using NEA-published emission factors, automatically updated in the platform. ▪ Scope 2 emissions calculated from electricity usage per asset/ system. • Benchmarking Tools ▪ Compare KPI performance across similar asset types, building typologies, or against Green Mark reference benchmarks.

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				• Reporting & Export ▪ Export charts and KPI data to PDF, XLS, or API for reporting to BCA, internal stakeholders, or ESG auditors. • Alert & Notification System ▪ Configurable thresholds trigger alerts when KPI exceeds set targets • Mobile/Tablet Responsive ▪ Dashboards are optimized for mobile and tablet use by FM teams on-the-go or in centralized control rooms.
	<u>Configurable Operations Dashboards</u>	<u>Create custom dashboarding and user journeys to suit various operations team requirements. Dashboards to provide operations team insights into systems operations</u>	<u>Optional</u>	• The Configurable Operations Dashboards function enables operations teams to create customized dashboards tailored to specific systems (e.g., ACMV, lighting, electrical), building zones, or user roles. These dashboards consolidate relevant real-time and historical data, providing actionable insights for day-to-day building operations, performance monitoring, and fault triaging. • Customizable Widgets ▪ Drag-and-drop interface for assembling dashboard components. ▪ Supports widgets like real-time values, trend graphs, alerts, control toggles, maps. • Role-Based Dashboard Views ▪ Custom dashboards based on user role (e.g., M&E supervisor, energy manager, technician). ▪ Each user sees only relevant system KPIs and alerts. • Multi-System Integration ▪ Displays data from ACMV, lighting, electrical, water, lift systems, access control, and fire systems. ▪ Capable to integrate with other 3rd party systems via BACnet, Modbus, OPC UA, MQTT. • Geospatial Layouts ▪ Interactive building/floor plans with clickable zones and systems. ▪ Supports status overlays, heatmaps, and zone-based alarm visualisation. • Real-Time & Historical Views ▪ Displays real-time system states and historical trends (hourly, daily, monthly). ▪ Includes options for zooming, time range comparison, and overlays. • System Insights Panel ▪ Summarizes asset status, alarms, efficiency levels, and runtime anomalies in an insight-driven layout. • Pre-Configured Templates ▪ Comes with templates for standard operational use cases (e.g., AHU monitoring, chiller plant status, lighting automation dashboard). • Alert & Fault Visualization ▪ Faults and alarms visually represented by severity and location. ▪ Enables quick triage of building-wide operational issues. • Mobile-Responsive Interface ▪ Accessible via tablets and smartphones, enabling field technicians to view dashboards while onsite. • User Journey Mapping ▪ Allows definition of operational workflows (e.g., start-of-day checks, energy reviews, fault response steps) within dashboards. • Data Export & Sharing ▪ Supports export to XLS/ PDF or API integration to share operational insights with external stakeholders or FM teams.
	<u>Certification Tracking</u>	<u>Utilise systems and utility data to report on compliance with various certifications. Develop additional reports as new certifications are</u>	<u>Optional</u>	• The Certification Tracking function provides visibility into compliance and progress towards sustainability and operations-related certifications (e.g., BCA Green Mark, SS 564, ISO 50001, GRESB, LEED). It leverages real-time systems and utility data to automate the

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		<u>targeted</u>		tracking of required performance indicators, generate supporting documentation, and streamline audits. • Certification Framework Integration ▪ Pre-loaded compliance templates for local and international standards: - o BCA Green Mark (Super Low Energy, Healthier Buildings, etc.) - o SS 564 (Energy Management) - o ISO 50001 (Energy Management Systems) - o GRESB, LEED, WELL, Fitwel • Automated Data Mapping ▪ Maps live BMS data streams (e.g., energy, water, indoor air quality) to relevant certification requirements. • KPI Tracking Dashboard ▪ Dashboards display real-time progress towards certification KPIs: - o EUI, WUI, chiller plant efficiency - o Carbon intensity, IAQ compliance • Audit Readiness Features ▪ Document repository for storing evidence such as calibration logs, maintenance records, IAQ sensor outputs. ▪ Tracks submission timelines. • Custom Report Builder ▪ Drag-and-drop report builder to create custom outputs aligned to certification forms or audits. • Alerting System ▪ Alerts for KPI non-compliance or certification milestone deadlines. • Export & Submission Tools ▪ Export reporting documents in formats accepted by certifying bodies (PDF, XLS, JSON). ▪ Supports automated submission to BCA portals (if enabled). • Adaptive Certification Templates ▪ Ability to onboard and configure new certification tracking templates as standards evolve (e.g., BCA Green Mark 2025 updates).
	<u>User Engagement</u>	<u>Provide departments and tenants with utility reporting to assist with emissions reporting</u>	<u>Optional</u>	• The User Engagement function enables departments, business units, or tenants to access tailored utility consumption and emissions reports. The system promotes sustainability awareness and supports scope-level carbon reporting by decentralising energy and resource performance data. • User/ Zone-Level Reporting ▪ Individual dashboards for tenants, departments, or functional zones. ▪ Custom permissions for viewing relevant utility and emissions data. • Utility Data Access ▪ Electricity, water, and gas consumption tracked by sub-meter or virtual allocation. ▪ Real-time and historical (hourly, daily, monthly). • Carbon Emission Display ▪ Scope 2 emissions automatically calculated from utility data using NEA-published grid emission factors. • Report Download & Sharing ▪ Users can download monthly or yearly summaries (PDF/XLS/CSV). ▪ Optional API integration for corporate sustainability or ESG reports. • Green Tips & Comparisons ▪ Display contextual comparisons (e.g., energy vs previous month, against building average). ▪ Optional behavioural nudges or alerts. • Mobile Dashboard Access

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				▪ Responsive interface for smartphones and tablets to allow anytime access. • Access Authentication ▪ Role-based access tied to user IDs, unit numbers, or department codes. Supports SSO, QR, or secure link login. • Compliance Readiness ▪ Supports Green Lease clauses and Scope 3 reporting obligations for tenants or departments.
7c	<u>Insight dashboarding</u>			
	<u>KPI tracking</u>		<u>Optional</u>	• This function provides real-time and historical performance dashboards for key BMS and sustainability KPIs, enabling stakeholders to make data-driven decisions. It supports energy optimisation, fault identification, and Green Mark compliance reporting through visual analytics aligned with Singapore's regulatory and climate context. • Real-Time Data Visualization – Dashboards update every 5–15 minutes with live BMS data (e.g., chiller efficiency, lighting load, AHU temperatures). • KPI Library ▪ Energy Utilisation Index (EUI) ▪ Water Use Intensity (WUI) ▪ Chiller plant efficiency (kW/RT) ▪ Airside performance (temperature delta, VAV control) ▪ Equipment uptime & fault trends ▪ Carbon Intensity (kgCO₂e/m²/year) ▪ Renewable energy contribution (%) • Customisable Views – User-configurable dashboards by role (Ops team, ESG lead, Facilities Manager). Site, zone, or portfolio-level toggles available. • Threshold Alerts & Colour Coding – Real-time flagging of KPIs breaching preset limits. Green/Amber/Red status indicators. • Historical Trend Analysis – Visual charts (line, heatmap, bar) with filters for daily, weekly, monthly, and annual periods. Option to overlay external data (weather, occupancy). • Benchmarking – Compare KPI performance across buildings or zones. Ranking table with best/worst performer flags. • Web Access – Fully responsive dashboard accessible via secure web portal. • Export Functionality – PDF, PNG, Excel exports for management reports and presentations. API access for data push to ESG platforms. • Green Mark Compliance – Real-time visibility of scoring KPIs (e.g., chiller system efficiency, renewable penetration). • ESG Reporting – Monthly KPI trends for Scope 2 emissions, water use, and system reliability for GRI/TCFD standards. • Operational Improvement – Enables proactive fault detection, energy-saving opportunities, and performance optimization. • Integration Points ▪ BMS: Primary data sources (BACnet, Modbus, OPC, MQTT) ▪ FM/CMMS Platforms: Link fault trends with maintenance history ▪ IoT Sensors: For granular metrics (CO₂, occupancy, lux levels, IAQ) ▪ Sustainability Dashboards: API integration with ESG reporting suites • Dashboard design complies with Green Mark: Performance Tracking (ENRB) and Smart FM Maturity Level 3–5 requirements. • Supports long-term digital transformation goals under BCA's Built Environment Industry Transformation Map (ITM).
	<u>Asset management</u>	<u>View all assets in one location to identify where resources are best spent to improve overall portfolio/fund performance and Leverage data to</u>	<u>Optional</u>	• The Asset Management Dashboard enables centralized visibility of all key mechanical, electrical, and environmental assets across a building or portfolio. This dashboard supports strategic resource planning by identifying performance issues, high-maintenance assets, and

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		<u>make informed decisions</u>		opportunities for efficiency improvements. - Unified Asset Registry - Central listing of all monitored assets (e.g., chillers, AHUs, pumps, UPS, lifts, FA panels). - Link with CMMS, BMS, and BIM systems. - Portfolio-Level Asset View - Multi-site visibility showing asset performance across buildings in a single view. - Health & Condition Metrics - Tracks operational parameters (e.g., runtime hours, vibration, temperature, pressure). - Highlights assets with frequent alerts or failures. - Asset Prioritization Scoring - Uses data (failure rate, energy use, lifecycle cost) to rank where maintenance or upgrades would yield highest impact. - Lifecycle Analysis - Dynamic end-of-life prediction based on asset condition data and original commissioning benchmarks. - Energy Impact Assessment - Shows each asset's energy contribution relative to total consumption - Maintenance Status Overlay - Visual tag for preventive, corrective, or predictive maintenance status. - Data Drill-Down - Clickable dashboards to navigate from building → system → asset → parameter level. - Interoperability - Integrates via APIs with CMMS, digital twin platforms, BMS (BACnet, OPC UA), and BIM models (IFC format). - Mobile Access - Field-ready dashboards viewable on tablets or smartphones, enabling on-site asset checks. - Reporting & Exporting - Export asset performance reports, asset condition summaries, or charts for reporting to sustainability committees or fund managers.
	Procurement Insights	<u>Leverage work order, asset, and operational data to identify equipment replacement strategies:</u> - Use performance data to inform CAPEX replacement strategies; - Inform future procurement strategies based on best manufacturers.	Optional	- To design, develop, and implement an interactive dashboard that leverages asset, work order, and performance data to inform equipment replacement strategies, capital expenditure (CAPEX) planning, and vendor performance benchmarking. - Asset Replacement Strategy - End-of-Life (EOL) Forecasting - Use asset metadata (install date, warranty period, average lifecycle) combined with usage and condition data to calculate estimated remaining useful life (RUL). - Provide alerts for assets nearing EOL thresholds based on configured rules. - Condition & Performance-Based Prioritization - Generate a prioritization score for replacement based on: - Frequency of failures (Mean Time Between Failures – MTBF) - Downtime duration (Mean Time to Repair – MTTR) - Energy efficiency degradation - Operational criticality (impact of failure on building performance) - Replacement Scenario Planning - Support scenario modeling based on budget constraints, building priority, or system criticality. - Enable users to simulate and compare different replacement schedules. - Visualize asset clusters for bulk procurement planning. - CAPEX Planning Insights - Multi-Year Budget Forecasting - Generate 3–10 year rolling CAPEX projections based on:

				- Replacement urgency scores - Equipment cost database (including historic procurement costs) - Inflation or escalation factors configurable by the user ▪ Risk-Based Prioritization - o Quantify asset risk and impact based on operational failures, energy wastage, and environmental impact (Green Mark indicators). - o Assign a weighted priority index for CAPEX approval workflows. ▪ Portfolio-Level Views - o Allow aggregation of CAPEX plans across multiple towers and departments. - o Enable filtering by towers, levels departments, and system type (e.g., HVAC, lifts, lighting), or asset manufacturer. • Vendor and Manufacturer Insights ▪ Supplier/Manufacturer Evaluation - o Track and compare: - Asset lifespan performance by manufacturer - Frequency and severity of failures post-installation - Response time and service quality by supplier ▪ Procurement Analytics - o Analyze historical procurement trends, including: - Cost variations by year and supplier - Lead times and delays - Warranty claims success rate ▪ Strategic Sourcing Insights - o Highlight consistently high-performing vendors for specific asset categories. - o Recommend preferred vendors based on performance-to-cost ratio. • Operational Analytics Module ▪ Real-Time Monitoring Integration - o Allow real-time asset health indicators from IoT/BMS integration (e.g., temperature, vibration, runtime hours). - o Enable streaming analytics for fault detection and early warnings. ▪ Maintenance Optimization - o Identify trends in corrective vs. preventive maintenance. - o Suggest optimization of PM schedules based on asset behavior and reliability data. ▪ Energy & Resource Efficiency - o Integrate energy and water usage data from BMS to track inefficiency linked to aging or underperforming assets. - o Quantify cost impact and environmental footprint of inefficient equipment. • User Interface & Usability Features ▪ Custom Dashboards: Allow role-based views for facility managers, finance teams, and procurement officers. ▪ Dynamic Filtering: Filter data by building, floor, asset category, vendor, system, or date. ▪ Drill-Down Navigation: Click from summary charts into detailed records (e.g., asset → work orders → vendor details). ▪ Export & Reporting: Export dashboard views and insights to Excel, PDF, and automated email reports. ▪ Mobile Responsiveness: Ensure compatibility on tablets and desktops for on-site inspections. • Advanced Capabilities ▪ AI/ML Predictive Modeling: - o Predictive failure models based on supervised learning - o Anomaly detection using unsupervised learning on sensor data ▪ Green Mark Integration:
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				- o Track Green Mark KPIs and tag assets based on environmental certification status ▪ Integration with BIM / Digital Twin: - o Linking insights to spatial visualizations (2D/3D) of buildings and systems
7d	<u>Centralized operations</u>			
	<u>Centralised Technical Operations</u>	<u>Utilize software tooling and reduce manpower by centralizing technical operations teams into one location. Remote visibility to all Operation Technology systems, predictive analysis and on-demand activations which can enhance workforce efficiency.</u>	<u>Optional</u>	• This function enables a centralised operations control centre for managing technical systems across one or multiple buildings. Leveraging integration with BMS and other Operational Technology systems, it provides real-time visibility, remote command capabilities, predictive analytics, and incident response automation—thereby reducing the need for on-site manpower and enhancing operational responsiveness. • Remote Systems Integration – Real-time connection to BMS, EMS, lighting control, ACMV systems, utility meters, lift monitoring, fire alarms, and IoT. • Unified Operations Dashboard – Web-based platform showing status of critical systems across all sites. Customisable views for power, HVAC, water, alarms, occupancy, and system health. • Remote Command Capability – Enable control actions (e.g., chiller reset, lighting zone off/on, AHU override, genset test mode) from a central location, with audit trails. • Predictive Analytics Engine – Uses machine learning models or rules-based analytics on historical and live data to detect anomalies, degradation trends, and fault precursors (e.g., rising condenser pressure, coil fouling, compressor cycling). • Workforce Scheduling & Activation – Integrated with CMMS or FM platform for on-demand task creation, smart routing of technicians, and exception-based responses. • Alert Consolidation – Alarm correlation logic to reduce noise and isolate root cause across buildings. Supports prioritisation based on impact and system criticality. • Failover & Resilience – Dual data paths with cloud and on-premise buffering to ensure system continuity. Role-based escalation procedures. • Compliance & Logging – Secure audit trails of all user actions, system events, override logs, and command history—meeting PDPA and ISO standards. • Integration Points ▪ BMS / Operational Technology Gateways: BACnet, Modbus, OPC-UA, MQTT protocols ▪ CMMS/ FM System: For task management, SLA monitoring, technician dispatch ▪ Security & Access Control Systems: For alarm monitoring and access audit ▪ Analytics Platforms: Cloud or local ML engines • Supports Singapore's Smart Estates framework and IMDA Smart FM Technology Roadmap • Aligned with Green Mark: Operations & Maintenance credits for smart control and manpower optimization
	<u>Reporting</u>	<u>Automated weather normalized benchmarking to report on portfolio initiatives. Use utility bills to show a 'before/after' state to validate project assumptions</u>	<u>Optional</u>	• This function enables automated benchmarking and performance tracking of buildings across a portfolio by normalising energy usage with local weather data (e.g., temperature, humidity). It validates the impact of energy efficiency projects using utility bill data to compare pre- and post-intervention performance, supporting ESG and Green Mark reporting. • Utility Bill Ingestion – Manual upload or automated collection (PDF, XLS, API-based from utility provider). Includes energy (kWh), water (m³), and gas usage. • Weather Data Integration – Pulls daily or hourly temperature and humidity data from Singapore's NEA or third-party APIs. Option to use building's own outdoor sensors if available. • Weather Normalization Algorithm – Applies regression analysis or degree-day method (Cooling Degree Days – CDD) to adjust energy usage based on ambient conditions. • Baseline Definition – Configurable baseline period (e.g., 12-month average pre-retrofit) with seasonal smoothing. • Before/After Project Comparison – Visual and tabular comparison of normalized vs actual energy/water use, CO₂ emissions, and cost savings. • Portfolio Benchmarking – Ranks multiple sites/buildings by normalized EUI, WEI, CO₂

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			- intensity, % savings. Enables best/worst performer tracking. - Project Validation Report – Auto-generated report for measurement & verification (M&V) showing project assumptions, targets, baseline, savings achieved, and weather adjustments. - Target Variance Alerts – Flags buildings deviating from expected normalized performance after intervention. - Data Confidence Scoring – Qualitative indicator based on data completeness, weather match quality, and utility bill consistency. - Green Mark Compliance – Supports GM: ENRB Part 5 (Performance Improvement Verification) and SLE certification documentation. - ESG Reporting – Enables energy/carbon intensity reporting per GRI 302 and TCFD, including impact tracking of portfolio-wide initiatives. - Output and Reporting<table><tr><th>Report Type</th><th>Contents</th><th>Frequency</th></tr><tr><td>M&V Validation Report</td><td>Weather-adjusted EUI trends, baseline vs post-retrofit, % savings</td><td>Monthly, Project Completion</td></tr><tr><td>Portfolio Benchmark Report</td><td>EUI, WEI, CO₂ intensity across buildings</td><td>Quarterly, Annually</td></tr><tr><td>Exception Reports</td><td>Buildings underperforming vs baseline or target</td><td>Monthly</td></tr></table> - Integration Points - BMS: Provides real-time and historical energy data. - Weather API or NEA Data Source: For outdoor temperature/humidity feeds. - CMMS / Project Tracker: Links completed retrofit or optimisation projects to actual performance outcomes. - ESG Dashboard: Normalized energy and emissions reporting for sustainability teams. - Based on Singapore's climate (tropical, consistent CDD year-round) – avoids over-correction common in temperate models. - Compliant with ISO 50006:2014 and supports BCA Super Low Energy building operational review processes.	Report Type	Contents	Frequency	M&V Validation Report	Weather-adjusted EUI trends, baseline vs post-retrofit, % savings	Monthly, Project Completion	Portfolio Benchmark Report	EUI, WEI, CO ₂ intensity across buildings	Quarterly, Annually	Exception Reports	Buildings underperforming vs baseline or target	Monthly
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<u>Avoid Unnecessary Callouts</u>	<u>Save costs and emissions by remotely triaging site issues to avoid unnecessary callouts. Utilize technical resources to diagnose faults prior to dispatching technicians</u>	<u>Optional</u>	- This function enables remote triaging of building system faults using real-time data and automated diagnostics to minimise unnecessary technician dispatches. It reduces operational carbon emissions, optimises manpower, and increases system uptime—supporting ESG goals and smart building outcomes. - Real-Time Fault Detection – Monitors BMS/EMS alerts from ACMV, lighting, lifts, pumps, electrical systems, and utility meters using predefined thresholds and rules (e.g., chiller temperature deviation, pressure anomalies). - Remote Diagnostics Dashboard – Interactive view showing current system status, historical trends, fault conditions, and suggested root causes. Accessible by facility and vendor teams. - Automated Triage Rules Engine – Configurable decision-tree logic or AI algorithms for fault categorization: Minor – Monitor, Moderate – Schedule visit, Critical – Immediate dispatch. - Live System Access – Enables authorised personnel to remotely connect to affected systems (via VPN or secure cloud gateway) for deeper checks. - Historical Fault Analysis – Trend analysis of fault patterns by system, zone, or asset type to identify recurring problems or false alarms. - Dispatch Recommendation – System flags whether a technician is needed, based on diagnostic result. Options: Remote Reset, Escalate, On-site Required. - Digital Work Order Integration – Automatically generates or suppresses work orders in the CMMS/FM system based on triage output. - Audit Trail – Full log of triage actions, overrides, recommendations, and resolution outcomes to support compliance and vendor accountability. - Integration Points												

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				▪ BMS Platforms: Ingests alarms and sensor data for diagnostics. ▪ CMMS/ FM Systems: Automates or suppresses job order creation. ▪ Remote Access Tools: Secure tunnel (e.g., VPN, RDP, VNC) for live system access. ▪ Data Lake/ Analytics Engine: Stores fault logs and supports advanced analytics. • Supports Green Mark for Existing Non-Residential Buildings (GM ENRB) criteria under operations & maintenance. • Aligns with IMDA Smart Facilities Management (SFM) frameworks and SS 599 standards on smart FM system interoperability.									
8	<u>Sustainability Management, Environmental, social and governance (ESG) reporting</u>	<u>- Captures data from energy, water and waste sectors, enabling real-time monitoring and management reporting.</u> <u>- Carbon capture insights will become mandatory in the future to all stakeholders.</u> <u>For example GreenMark report:</u> <u>- Building energy efficiency performance indicators</u> <u>- Total active energy (kWh) and total active power (kW)</u> <u>- Total active energy by</u> <u>For example ESG report:</u> <u>- Energy utilisation index (EUI) and water efficiency index (WEI) by month and year</u> <u>- Total recycling rate by month and year by month and year</u> <u>- EUI breakdown</u> <u>- Total energy trends</u> <u>For example Water Metering:</u> <u>- Water use intensity (L/m2), Water consumption summary table (last 24hrs, week, month and year)</u> <u>- Water consumption (L) by date and category</u>	<u>Optional</u>	• The system is intended to support compliance with BCA Green Mark, PUB, NEA, and global ESG frameworks including Global Reporting Initiative (GRI) and Task Force on Climate-related Financial Disclosures (TCFD). • The system shall capture, processes, and reports data across energy, water, and waste sectors. It enables real-time monitoring, analytics, and reporting of sustainability performance, including future-ready carbon emission insights. • System Function Modules: ▪ Energy Meters: Total energy and power readings by system and source ▪ Water Meters: Potable, NEWater, cooling tower, and irrigation ▪ Waste Monitoring: Categorised by general, recyclable, and food waste ▪ Carbon Factors: Localised (NEA-published) emission coefficients ▪ APIs/IoT Integrations: Real-time data feeds from BMS, EMS, FM platforms ▪ Time-series aggregation (15-min, hourly, daily, monthly, annually) ▪ Benchmarking and baseline tracking ▪ KPI computation (e.g., EUI, WUI, recycling rate) ▪ Anomaly detection and threshold alerting • Energy Monitoring ▪ Total Active Energy (kWh): 15-min granularity ▪ Total Active Power (kW): Instantaneous; logged at 5-15 min intervals ▪ Energy Utilisation Index (EUI): kWh/m²/year, system-wise breakdown ▪ Green Mark Indicators: Energy savings vs baseline, chiller efficiency ▪ Trend Dashboard: Daily/monthly/yearly trends; target comparisons ▪ Alerting System: Peak load exceedance, inefficiency flags • Water Monitoring ▪ Water Use Intensity (WUI): Litres/m²/year ▪ Real-Time Usage: Data every 15 minutes ▪ Consumption Summary Table: Data for last 24 hrs, week, month, year ▪ Water Efficiency Index (WEI): Monthly/yearly benchmarking ▪ Leak Detection: Alerts for abnormal consumption patterns • Waste Monitoring ▪ Waste Volume/Weight: Categorised data input (kg or tonnes) ▪ Recycling Rate: Monthly/annual; % recyclables of total waste ▪ Waste Trends: Category-wise trend visualisations ▪ Manual Data Portal: Upload interface for external waste contractor data ▪ Compliance Tags: Hazardous/regulated waste annotation • Carbon & ESG Reporting ▪ Carbon Emission Calculation: Scope 1 & 2 using NEA grid factors ▪ GHG Dashboard: Emission breakdown by system and source ▪ Report Generation: ESG and Green Mark templates ▪ Audit Trail: Logging for all manual inputs and overrides • Output and Reporting<table><tr><th>Report Type</th><th>Format</th><th>Frequency</th></tr><tr><td>Green Mark Report</td><td>PDF/XLS (BCA standard)</td><td>Quarterly, Annually</td></tr><tr><td>ESG Report</td><td>PDF/XLS/JSON (GRI/TCFD)</td><td>Monthly, Annually</td></tr></table>	Report Type	Format	Frequency	Green Mark Report	PDF/XLS (BCA standard)	Quarterly, Annually	ESG Report	PDF/XLS/JSON (GRI/TCFD)	Monthly, Annually
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				Dashboard Export API Integration formats) PNG/PDF JSON/RESTful On-demand Real-time or batch - Future-Readiness - Carbon Capture Integration: Design for forward compatibility - AI-Based Analytics: Forecasting, anomaly prediction - Smart City Data Exchange: Enable linkages to national sustainability platforms												
9	Digital Twin & Occupancy Data	- Leverage granular data to provide occupancy data - Deploy data driven solution on actual occupancy and not using estimated numbers	Optional	- This module captures and analyses real-time and historical occupancy data across the building to support intelligent space utilization, ESG reporting, and operational efficiency. Unlike traditional static occupancy models, this system uses real sensor inputs to reflect actual building usage, in line with Singapore's growing emphasis on smart facility management and demand-side efficiency. - Occupancy Data Capture – Integration with smart sensors (e.g., PIR, CO₂, BLE, Wi-Fi tracking, camera-based analytics). Sensors deployed in key zones: meeting rooms, open office spaces, washrooms, lift lobbies, carparks, and retail areas. - Digital Twin Engine – Real-time virtual representation of spaces, updated via API and IoT feeds. Supports 2D/3D overlays and BIM-based mapping. - Data Granularity – Minimum 1-minute intervals; adjustable to 15-minute aggregation for reporting. - Zone-based Occupancy Metrics – Headcount, utilization rate (%), time of use, dwell time. - Occupancy Heatmaps – Visual representation of density and usage trends by zone and time (e.g., peak/off-peak periods). - ESG Integration – Occupancy-adjusted energy intensity (kWh/person) and carbon intensity (kgCO₂e/person). - Smart Scheduling – Optimised lighting and HVAC operation based on real occupancy, reducing energy waste in under-utilised zones. - Data Accuracy – System design to meet minimum 85–95% detection accuracy depending on sensor type and environment. - Privacy Compliance – Anonymised data aggregation, camera feeds (if used) processed at edge with no facial recognition. In compliance with Singapore's PDPA. - Output and Reporting<table><tr><th>Report Type</th><th>Contents</th><th>Frequency</th></tr><tr><td>Occupancy Utilisation Report</td><td>Daily/hourly usage % by zone, peak occupancy times</td><td>Daily, Weekly</td></tr><tr><td>Digital Twin Activity Log</td><td>Log of space usage changes and anomalies</td><td>Real-time, Monthly</td></tr><tr><td>ESG Occupancy Metrics</td><td>Energy use per occupant, CO₂/person, space efficiency</td><td>Monthly, Annually</td></tr></table> - Integration Points - BMS: To enable adaptive HVAC and lighting based on occupancy. - Facility Management Platform: For room booking, space planning, maintenance scheduling. - Sustainability Reporting Dashboard: To enrich EUI and carbon intensity with real occupant load. - Supports Green Mark Super Low Energy (SLE) targets by reducing wastage through demand-based control. - Aligns with IMDA's Smart Estates and BCA's digital delivery roadmap, facilitating digital twin adoption	Report Type	Contents	Frequency	Occupancy Utilisation Report	Daily/hourly usage % by zone, peak occupancy times	Daily, Weekly	Digital Twin Activity Log	Log of space usage changes and anomalies	Real-time, Monthly	ESG Occupancy Metrics	Energy use per occupant, CO ₂ /person, space efficiency	Monthly, Annually
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10	Asset Performance Management	IoT sensors to monitor MEP system. For example: - IoT water level sensor in Inspection Chamber (IC) pits waste water level and remotely trigger	Base	To implement an IoT-enabled Asset Performance Management (APM) system that integrates with the existing Integrated Building Management System (IBMS) to enhance real-time monitoring, predictive maintenance, and fault detection of critical MEP systems.												

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		<u>alert to IBMS for FM to respond when IC pit is choked with debris.</u> <u>- IoT vibration and temperature sensors on AHU, Cooling Tower, pumps and other mechanical system motors.</u>		- The system shall enable real-time monitoring, fault detection, and predictive maintenance of MEP equipment, including but not limited to Inspection Chamber (IC) pits, AHUs, cooling towers, and pumps. - Utilising the sensors provisioned and installed in base tender, the APM shall: - Support real-time monitoring, historical data trending, and rule-based alerting - Be accessible via a secured dashboard - Support support predictive maintenance with AI/ ML - Integrate with sustainability dashboards for ESG reporting
	<u>Asset Management</u>	<u>Register assets both robots and physical hard assets on the platform to track for preventive and predictive analysis.</u>	<u>Optional</u>	- To establish a centralised digital registry and tracking system for both physical and robotic assets. This enables: - Visibility of asset inventory, deployment location, and operational status; - Lifecycle management, from installation and warranty tracking to decommissioning and asset disposal; - Preventive maintenance planning, through scheduled servicing based on time or usage thresholds; - Predictive maintenance, using IoT sensor data and analytics to forecast failures before they occur; - Integration with facility management (FM) workflows, automating task assignments, alerts, and compliance checks; - Support for Green Mark Smart FM and digital twin initiatives by maintaining a digital record of physical assets and their performance over time. - Functionality: Ability to register, categorize, and monitor assets (both robotic and physical hard assets) - Asset Types Supported: Mechanical (e.g., pumps, AHUs), Electrical (e.g., switchgear), Plumbing, Robotic (e.g., cleaning robots, security droids) - Registration Capability: Manual and automated registration via API, barcode, QR code, or NFC tagging - Asset Data Fields: Asset ID, Type, Location, Manufacturer, Model, Serial Number, Installation Date, Warranty Info, Maintenance Schedule - Data Sources: Integration with IoT sensors, BMS systems, CMMS, and robotic system controllers - Communication Protocols: REST API, MQTT, BACnet/IP, Modbus TCP - Predictive Analytics: AI/ ML engine for trend detection, fault forecasting, and maintenance optimization - Preventive Maintenance: Rule-based scheduling, alerting, and task generation based on usage data - Dashboard: Centralized FM dashboard with asset tree view, health status, and KPIs - Compliance: BCA Green Mark Smart FM Framework, PDPA for personal/ operational data, IT Cyber Security - Security: Role-based access control, audit logs, TLS encryption for data in transit - Mobile Access: Support for mobile FM apps for asset scanning and update in the field - Bi-directional data sync with BMS/IBMS for real-time health and operational status - All registered assets must be viewable in the IBMS or FM dashboard with status indicators and filter/ search functionality. - The system shall support lifecycle tracking from commissioning to decommissioning. - Robotic assets (e.g., autonomous cleaners or delivery bots) shall include integration with movement logs, maintenance cycles, and battery performance metrics. - Physical hard assets must support maintenance logging and compliance audit trail. - Support for alerts, condition monitoring, and preventive action automation must be included.
	<u>Maintenance Management</u>	<u>Capture information related to routine and scheduled maintenance for robots, sensors and</u>	<u>Optional</u>	The proposed Maintenance Management System (MMS) will automate and optimize the management of routine, scheduled, and ad-hoc maintenance tasks for robots, sensors, and

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	<u>physical assets.</u> <u>Automate generating Work Orders and tracking.</u> <u>For example:</u> - <u>Work order response trend</u> - <u>Work order rectification trend</u> - <u>Most common work order tasks</u> - <u>Top work order types</u> - <u>Work order priority breakdown</u> <u>Provides a checklist for maintenance and users can document ad-hoc repair tasks. For example:</u> - <u>Scheduled Audit ratio (Schedule, Completion, Outstanding, Upcoming)</u> - <u>Scheduled Maintenance (Schedule, Completion, Outstanding, Upcoming)</u> <u>Maintenance feedback tracking from users. For example:</u> - <u>Feedback work order completion</u> - <u>Total negative feedback received</u> - <u>Completed feedback work order</u> - <u>Negative feedback summary</u> - <u>Total feedback by location and month</u>	physical assets in a Building Management System (BMS). It will integrate with existing BMS hardware and software, capturing critical maintenance data, generating work orders, and enabling real-time tracking and feedback mechanisms. • Routine and Scheduled Maintenance ▪ Maintenance Work Order Generation: - o Automated Creation: Generate work orders automatically based on predefined schedules or system alerts from robots, sensors, and physical assets. - o Scheduling Criteria: Include parameters such as asset life cycle, maintenance type (preventative, corrective), asset performance metrics, and compliance with local regulations (e.g., periodic checks for fire safety systems, air conditioning servicing). - o Work Order Attributes: Each work order should include: - Work Order ID, Type, Priority, Description, Assigned Personnel, Expected Completion Date, Location, Assets involved, Estimated Cost, and Materials required. ▪ Work Order Response and Rectification Trend: - o Track the time taken from work order creation to response. - o Track the time taken for the rectification of issues identified in work orders. - o Statistical reporting to generate trends in response and rectification times for performance analysis. ▪ Common Work Order Tasks and Types: - o Categorize work orders based on tasks (e.g., sensor calibration, cleaning, repair, inspection). - o Analyze and report on the most common types of tasks (e.g., cleaning, air filter changes) to forecast labor and material requirements. ▪ Work Order Priority Breakdown: - o Work orders are classified into priority levels (e.g., Critical, High, Medium, Low). - o The system will track the breakdown of work orders by priority and highlight overdue critical tasks. • Maintenance Checklist and Ad-Hoc Repair Tasks ▪ Routine Maintenance Checklist: - o Provide an interactive digital checklist for maintenance staff to complete tasks as per the scheduled maintenance. - o Items to include: visual inspections, operational checks, and minor corrective actions. - o Checklist will be customizable based on asset type, regulatory requirements, and maintenance guidelines. ▪ Ad-Hoc Repairs: - o Function to document unscheduled repairs identified during regular operation or reported by BMS sensors or personnel. - o Work orders generated for these ad-hoc tasks will have an urgency level and can be escalated based on asset criticality. • Scheduled Audit and Maintenance Tracking ▪ Audit Ratio Tracking: - o Track the ratio of scheduled maintenance work completed against those planned (i.e., Completed, Outstanding, Upcoming). - o Include graphical and tabular representations of data to monitor maintenance compliance. ▪ Scheduled Maintenance Tracking: - o Real-time tracking of scheduled maintenance tasks to ensure they are completed on time. - o Visual reports on overdue tasks, upcoming tasks, and completion status. • Maintenance Feedback System ▪ User Feedback:
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				- o Allow users (maintenance staff, asset managers, or tenants) to provide feedback on work order completion. - o Include fields for rating the work quality, timeliness, and any issues encountered during the maintenance task. - o Categorize feedback into positive, neutral, or negative. ▪ Negative Feedback Management: - o Track total negative feedback received, providing insights into recurring issues or areas of concern. - o Summarize feedback with a severity classification and categorize by asset type or maintenance task. ▪ Feedback by Location and Month: - o Collect feedback data by location within the building, allowing for location-based analysis of maintenance performance. - o Monthly summary reports to track feedback trends and identify patterns. • Integration and Data Flow ▪ Building Management System (BMS) Integration: - o The MMS will integrate seamlessly with the BMS platform. APIs or middleware solutions will be used for real-time data exchange (e.g., sensor status, asset health). - o Integration with asset management databases to capture asset-specific data (e.g., asset ID, warranty status, inspection history). ▪ Maintenance and Work Order Databases: - o Centralized database for storing work orders, maintenance history, feedback data, and asset-specific records. - o Cloud-based or on-premise storage options based on system architecture preferences. • User Interfaces ▪ Web and Mobile Interfaces: - o A responsive web portal for real-time updates, work order creation, checklist management, and feedback submission. - o Maintenance staff can access work orders, checklists, and submit ad-hoc repair requests from any location within the building. ▪ Dashboard for Management: - o A comprehensive dashboard showing real-time maintenance data, trends, and KPIs (e.g., number of open work orders, feedback statistics). - o Role-based access control for different user levels (e.g., Admin, Maintenance Staff, Asset Managers) • Notification and Alerts System ▪ Automated Notifications: - o Automatic notifications via email/SMS for upcoming scheduled maintenance, overdue tasks, and work order completion. - o Alert system for critical tasks requiring immediate attention (e.g., sensor failure, fire alarm inspection). • Reporting and Analytics ▪ Work Order Analytics: - o Real-time reporting tools to generate trends on work order completion times, rectification rates, most frequent work order types, and priorities. - o Exportable reports (PDF, Excel) for review and analysis. ▪ Feedback Analytics: - o Analysis of maintenance feedback trends, broken down by location, asset type, and task type. - o Visual reports (graphs, heatmaps) for management to identify areas for improvement.
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	<u>Contractor Management</u>	- Capture all contractors expiry and renewal to prepare well ahead of time. - Performance of contractors can be tracked.	<u>Optional</u>	- The Contractor Management System (CMS) will be integrated into the Building Management System (BMS) to effectively track the performance, expiry, and renewal of contractor agreements for maintenance tasks and services. - Contractor Expiry and Renewal Management - Contractor Record Management: - Capture contractor details including name, company information, license number, services provided, and contract start/end dates. - Include data fields for regulatory compliance documents such as certifications, insurance, and licensing requirements specific to Singapore's building and construction regulations (e.g., BCA, MOM, or NEA compliance). - Contract Expiry Notifications: - Automated reminders and notifications sent well in advance of contractor license expiry or contract end date (e.g., 3 months, 1 month, and 1 week before expiry). - Notifications via email/SMS to relevant stakeholders (e.g., facility manager, procurement department). - Renewal Tracking: - Track the status of renewal applications, allowing users to monitor if the contractor has submitted renewal documents and if they are in the process of being reviewed. - Ability to track renewal deadlines and keep records of the renewal process to ensure timely contract continuity. - Document Management: - Integration of digital document management for contractor agreements, including scanned copies of signed contracts, insurance certificates, and compliance documentation. - Option to upload and link renewal documents directly to contractor profiles for easy access and audit purposes. - Contractor Performance Tracking - Performance Metrics: - Define and track key performance indicators (KPIs) for contractor performance, including: - Timeliness of work completion - Work quality and compliance with standards - Responsiveness to emergencies or work orders - Safety record (e.g., accident reports, safety audits) - Cost efficiency and adherence to budget - Customer feedback or satisfaction (e.g., feedback from maintenance staff or asset managers) - Performance Rating System: - A rating system where contractors are evaluated based on their performance for each task or service provided. - Ratings can be based on predefined metrics (e.g., Excellent, Good, Fair, Poor) or a numeric scale (e.g., 1 to 5). - Periodic performance assessments (quarterly, annually) based on collected data, feedback, and system-generated reports. - Contractor Performance History: - The system will maintain a performance history for each contractor, allowing for easy reference and trend analysis over time. - Reports summarizing historical performance, including areas of improvement or recurring issues (e.g., missed deadlines, subpar quality). - Dispute Management and Incident Reporting: - Functionality for logging incidents, complaints, and disputes related to contractors. - A system for tracking resolutions and action steps taken, including any corrective
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				actions or penalties. • Reporting and Analytics ▪ Performance Dashboards: - o Visual dashboards for managers to review contractor performance metrics in real-time. - o Provide breakdowns by contractor, service type, and KPI. - o Trending of contractor performance over time (e.g., service improvement or decline). ▪ Contract Expiry and Renewal Reports: - o Automated reporting on contractor contract expiry dates, renewal statuses, and compliance documentation. - o Report summaries for contractors that are nearing expiry and require immediate action for renewal. ▪ Compliance Tracking: - o Reports for compliance with local regulations (e.g., BCA licensing, safety compliance). - o Visual and tabular summaries of contractor compliance status for audit purposes. • Integration and Data Flow ▪ Building Management System (BMS) Integration: - o The CMS will integrate with the existing BMS, enabling the importation of contractor details and contract expiration data. - o Integration with procurement systems to facilitate contract renewal and management workflows. ▪ Contractor Database: - o A centralized database for storing contractor records, performance data, and historical information. - o Relationship mapping between contractors and their respective contracts, work orders, or assets they maintain. • User Interfaces ▪ Web and Mobile Interfaces: - o A web portal for contractor management, performance tracking, and notifications. - o User interfaces will allow authorized personnel (e.g., facility managers, procurement officers) to add, update, and review contractor details. ▪ Contractor Portal: - o Contractors can access a self-service portal to submit documents, track their performance ratings, and receive notifications regarding contract expiry, renewals, and performance feedback. • Notifications and Alerts ▪ Automated Expiry Notifications: - o The system will send automatic notifications when a contractor's contract or compliance certification is approaching expiry (e.g., 3 months, 1 month, 1 week). - o Alerts for upcoming renewals to remind both contractors and management teams to initiate the renewal process. ▪ Performance Alerts: - o Alerts for any negative performance reviews or incidents reported regarding contractors (e.g., safety breaches, missed deadlines). ▪ Escalation Mechanism: - o If renewal or compliance issues are not addressed in time, escalation alerts will notify higher-level management. • Reporting and Analytics ▪ Contractor Performance Reports: - o Monthly, quarterly, and annual reports summarizing contractor performance, categorized by service type and KPI metrics.
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				- o Ability to export reports to Excel, PDF, or CSV for review and analysis. ▪ Contract Expiry and Renewal Reports: - o Reports that display the contractor contract expiry timeline, renewal progress, and upcoming deadlines. - o Customizable filters for generating contractor-specific or company-wide renewal status reports. ▪ Feedback and Incident Reports: - o Reports summarizing user feedback on contractors and any incidents related to contractor performance. - o Categorization of feedback into positive, neutral, or negative for trend analysis.
	Alerts & Notifications	<u>Automate workflow processes across systems.</u> <u>Send an alert or trigger a work order for corrective action</u> <u>- Alerts can be displayed in Platform, via email or SMS, and can trigger Work Orders</u> <u>- Pre-empt potential failure of critical equipment and reduce time for operations teams to manually raise work orders</u> <u>- Provide checklist for maintenance and users can document ad-hoc repair tasks</u>	Optional	• The Alerts & Notifications function in the Building Management System (BMS) will automate workflow processes and streamline maintenance management by sending alerts across multiple channels (platform, email, SMS). This system will reduce manual intervention by triggering work orders for corrective actions and enabling proactive responses to critical equipment failures. • Work Order Automation ▪ Work Order Generation: - o Upon receiving an alert, the system will automatically generate a work order for corrective action. - o The work order will include: - Work Order ID - Asset involved and specific issue - Assigned technician or contractor - Priority level (based on the severity of the alert) - Estimated time of completion - Any necessary parts or materials - Relevant documents (e.g., manuals, schematics) ▪ Escalation Workflow: - o If a work order is not responded to within a predefined time, an escalation alert will be triggered to higher management or the next-level technician to ensure timely resolution. - o Work order statuses (e.g., Pending, In Progress, Completed) will be tracked in real-time to ensure accountability. ▪ Preventative Maintenance Scheduling: - o Alerts can also trigger preventative maintenance work orders based on equipment usage, historical failure trends, or manufacturer guidelines (e.g., time-based or condition-based maintenance schedules). ▪ Integration with Other Systems: - o The alerts and work order systems will integrate with other BMS components (e.g., Asset Management System, CMMS) to ensure data consistency across systems and seamless workflows. • Ad-Hoc Repair Tasks and Maintenance Checklist ▪ Ad-Hoc Repair Logging: - o Maintenance staff will be able to log ad-hoc repair tasks that occur during routine inspections or as identified by alerts. - o A form for ad-hoc repairs will be available within the platform, capturing essential data such as: - Asset involved - Issue description - Actions taken - Time spent on repairs - Parts replaced (if applicable)

				- Comments or additional notes ▪ Maintenance Checklist: - o The system will provide a digital checklist for scheduled and routine maintenance tasks (e.g., filter replacement, lubrication, system calibration). - o The checklist will be customizable based on asset type, regulatory requirements, and operational needs. - o The checklist will be updated in real-time as maintenance staff mark tasks as complete, providing immediate feedback on progress. ▪ Task Documentation: - o Maintenance staff can document and track the progress of maintenance and repairs via the checklist and work orders. - o Any ad-hoc or emergency maintenance activities will be documented in a log for later review, enabling trend analysis and continuous improvement. • Proactive Monitoring and Failure Prevention ▪ Predictive Maintenance Alerts: - o The system will use historical data, machine learning models, and trend analysis to predict potential equipment failures before they occur. - o For instance, based on factors like vibration patterns, temperature readings, or sensor data, the system will generate predictive alerts for equipment that might fail soon, triggering a preventive work order or inspection. ▪ Condition-Based Monitoring: - o Critical systems or assets will be continuously monitored for key performance indicators (KPIs), such as: - Power consumption - Temperature and humidity levels - Pressure readings - System uptime and performance metrics ▪ When these KPIs fall outside predefined ranges, the system will automatically alert the relevant team, triggering proactive maintenance. • Feedback and System Optimization ▪ Alert Feedback Mechanism: - o After resolving an alert, users (e.g., maintenance staff or facility managers) will be able to provide feedback on the effectiveness of the resolution and whether the corrective action fully addressed the issue. - o Feedback will include: - Resolution effectiveness (e.g., Fixed, Partial, Not Fixed) - Any ongoing issues or recurrence of the problem - Suggested improvements for system monitoring or alert thresholds ▪ Alert History and Analytics: - o Historical data on alerts and work orders will be stored and available for trend analysis. - o Analytics will provide insights into common equipment failures, recurring issues, and patterns over time, which will inform the maintenance strategy and optimize future operations. - o Reports will categorize alerts by asset, frequency, severity, and response time. • Integration and Data Flow ▪ BMS Integration: - o The Alerts & Notifications system will integrate with various BMS subsystems, including asset management, sensor monitoring, and work order systems. - o APIs or middleware will be used for seamless data exchange between systems, ensuring that alerts and work orders are generated in real-time. ▪ Sensor Data Integration:
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				- o The system will receive and process data from connected sensors (e.g., temperature, pressure, vibration, humidity) to detect deviations and trigger alerts automatically. - o Integration with IoT devices (e.g., smart thermostats, predictive maintenance sensors) to gather real-time data. • User Interfaces ▪ Centralized Dashboard: - o A web-based dashboard will provide a centralized view of all active alerts, work orders, and maintenance status in real-time. - o Alerts will be color-coded by severity (e.g., Red for critical, Yellow for high priority) to provide quick visual identification of urgent issues. • Notification Delivery ▪ Email and SMS Services: - o The system will use an integrated email and SMS gateway to send notifications to stakeholders, contractors, or external vendors. - o Template-based messages will ensure consistency in the format and information included in notifications. • Reporting and Analytics ▪ Alert and Work Order Analytics: - o Analytics tools will provide insights into alert frequency, work order completion times, and the efficiency of the response teams. - o Trend analysis will help optimize the system by identifying common failures and areas for improvement. ▪ Compliance Reports: - o Reports will be generated to ensure that all alerts and work orders comply with regulatory standards, including local building and safety regulations in Singapore (e.g., BCA, NEA).
11	<u>Smart Water Management</u>	<u>Leak detection systems: Sensors can detect leaks or unusual water flow patterns to prevent wastage and damage.</u>	<u>Base</u>	• The Smart Water Management function will utilize advanced leak detection sensors and monitoring technologies to identify and respond to water leaks or unusual flow patterns within the building's plumbing and water systems. This feature will integrate seamlessly with the BMS to prevent water wastage, minimize potential damage, and improve resource efficiency. • Leak Detection Systems ▪ Sensor Integration: - o Flow Sensors: These will measure water flow in pipes, tanks, and other critical water systems. They will detect deviations from normal flow patterns (e.g., continuous flow when the system is supposed to be idle). - o Pressure Sensors: Pressure sensors will monitor the pressure within the water distribution system to detect unusual drops or spikes that could indicate a leak or blockage. - o Moisture Sensors: Sensors placed in areas prone to water accumulation (e.g., under sinks, near water pumps, along pipe routes) will monitor moisture levels to detect early-stage leaks. - o Ultrasonic Sensors: Used to measure water flow and detect anomalies in the flow rate, helping identify hidden leaks within walls, ceilings, or underground pipes. - o Acoustic Sensors: These sensors use sound waves to detect abnormal noises that could signal leaks, such as water escaping from pipes. ▪ Leak Detection Algorithms: - o The system will analyze data from sensors to determine whether the readings fall outside expected parameters (e.g., continuous flow, pressure drop). - o It will use predefined thresholds based on building layout, water system characteristics, and historical usage patterns to trigger alerts when deviations are detected. ▪ Real-Time Data Monitoring:

				- o Sensors will provide real-time data on water flow, pressure, and moisture levels, enabling continuous monitoring of the system. - o Data will be transmitted to the BMS for processing, alerting, and decision-making. • Alerting and Notification ▪ Alert Types: - o Leak Detection Alerts: These will be generated when a water leak or unusual water flow pattern is detected. Alerts will be prioritized based on the severity and location of the issue. - o Water Waste Alerts: If the system detects unusual or excessive water flow (e.g., when water should not be flowing), it will trigger a water wastage alert to prevent resource loss. - o Low Pressure Alerts: In case of a pressure drop indicating potential leakage or a blockage in the system, an alert will be triggered to address the issue before it worsens. ▪ Alert Channels: - o Platform Notifications: Alerts will be displayed on the BMS dashboard for immediate action by facility management. - o Email and SMS Notifications: Immediate email and SMS alerts will be sent to designated personnel (e.g., building managers, maintenance teams) to ensure quick response times. - o Mobile Alerts: Maintenance teams will receive push notifications on their mobile devices for on-the-go awareness of any detected leaks or irregularities. ▪ Alert Severity and Escalation: - o Alerts will be categorized by severity (e.g., Critical, High, Medium, Low), ensuring that the most urgent issues are addressed first. - o For unresolved issues or delayed responses, an escalation protocol will notify higher-level management for further action. • Automatic Water Shut-Off ▪ Automated Shutoff Valve Integration: - o For high-severity leak alerts, the system can integrate with automated water shut-off valves that can be triggered to stop the flow of water to the affected area. - o This will minimize potential water damage and wastage, particularly in areas where leaks could cause significant damage (e.g., server rooms, bathrooms, kitchens). ▪ Remote Control: - o Facility managers or authorized personnel will have the ability to remotely control water shut-off valves through the BMS interface or mobile, allowing for manual intervention if necessary. • Data Logging and Reporting ▪ Leak History Logging: - o All leak events and water flow anomalies will be logged with timestamps, locations, and sensor data for future reference and analysis. - o This history will be accessible via the BMS dashboard for analysis, helping to identify patterns and recurring issues. ▪ Reports and Analytics: - o Detailed reports will be generated based on leak detection data, including: - Frequency and severity of detected leaks - Locations of most frequent leaks - Response times to detected issues - Water wastage volumes and trends - o Analytics will help optimize water management strategies and guide future building maintenance efforts. • Preventive Maintenance and System Health Monitoring
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				▪ Predictive Maintenance: ○ The system will track sensor health and water system performance over time to predict potential equipment failures (e.g., sensor malfunctions, valve issues). ○ Predictive alerts will notify facility managers when sensors or water system components require maintenance or calibration to avoid future issues. ▪ Routine Maintenance Scheduling: ○ Based on sensor data and system performance trends, the system will recommend and schedule routine maintenance tasks, such as cleaning or replacing sensors, checking valves, and inspecting water pipes. • Integration and Data Flow ▪ BMS Integration: ○ The leak detection system will integrate seamlessly with the existing BMS, ensuring that all data flows into a centralized platform for real-time monitoring and decision-making. ○ The BMS will manage the alerts, work orders, and notifications triggered by leak events and irregular water flow patterns. • User Interfaces ▪ Centralized Dashboard: ○ A web-based and mobile-friendly dashboard will provide an overview of the building's water system, displaying real-time data from sensors and alerts. ○ The dashboard will feature an interactive map of the building, highlighting areas where leaks or anomalies have been detected. • Reporting and Analytics ▪ Customizable Reports: ○ Reports can be generated on-demand or automatically at predefined intervals (e.g., weekly, monthly, quarterly) and can include data such as: - Leak detection and resolution times - Water wastage volumes and trends - System performance and sensor status ○ The system will support exporting data to various formats, including Excel, PDF, and CSV, for further analysis.
12	<u>Smart integration platform</u>	<u>On-Premise Production Servers Hosting, I/O Gateways, Database servers, etc</u>	<u>Base</u>	• The Smart Integration Platform will serve as the core infrastructure for hosting and managing on-premise production servers, I/O gateways, database servers, and other critical components required to support the BMS. It will integrate various systems (e.g., HVAC, lighting, security, energy management) within the building environment, ensuring seamless data exchange and efficient operation of all connected assets. • The platform will leverage on-premise hardware resources to manage and process large amounts of data in real-time, ensuring that critical systems like sensors, controllers, and other BMS subsystems are monitored, controlled, and optimized continuously. • On-Premise Production Servers Hosting ▪ Server Infrastructure: ○ On-premise production servers will host core BMS applications, services, and databases, ensuring that mission-critical data processing and operations occur within the building's local environment, rather than relying on cloud services. ○ Servers will support applications such as: - BMS central control software - Data analytics and reporting systems - Energy monitoring and optimization tools - Building automation system control logic - Predictive maintenance algorithms ○ The hardware provisioned shall be as per technical specifications. • Database Servers

				▪ Database Design and Architecture: ○ The database servers will store large volumes of real-time data, historical logs, and configuration settings for the BMS system. A high-performance relational database system (e.g., MySQL, PostgreSQL, or Microsoft SQL Server) will be used to handle structured data from BMS sensors and applications. ○ A time-series database (e.g., InfluxDB, TimescaleDB) will also be integrated for efficient storage and retrieval of time-stamped data such as temperature, pressure, or flow readings. ▪ Data Redundancy and Backup: ○ The database will be configured with replication, clustering, or sharding to ensure high availability, data redundancy, and fault tolerance. ○ Automated backup solutions will be implemented to back up critical data regularly (e.g., daily incremental backups, weekly full backups) to prevent data loss. ▪ Data Security: ○ Encryption: All sensitive data will be encrypted both at rest (in databases) and in transit (via secure protocols such as HTTPS, TLS). ○ Access Control: Role-based access control (RBAC) will be implemented to restrict database access to authorized users only, with strict logging and monitoring of database activity. ○ Audit Trails: The system will maintain audit logs to track all user activities (e.g., data access, changes to settings or configurations) for security and compliance purposes. ▪ Database Maintenance: ○ Automated database maintenance tasks such as index optimization, cleanup of historical data, and performance tuning will be configured to ensure optimal database performance.
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